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Project-Based Learning

Full Stack Development – BCS406

Project Report - Review 1

on

"TINY CARE-Integrated Child Health and Nutrition Monitoring System"

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Submitted By

Khushi C Hiremath 1DT24CS125
Lakshya D Basur 1DT24CS132
Namitha A Byrav 1DT24CS168

Under the Guidance of

Divya H N
Assistant Professor, Department of CSE

Department of Computer Science and Engineering

DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT
(Affiliated to VTU, Belagavi and Approved by AICTE, New Delhi),
CE, CSE, ECE, EEE, ISE, ME Courses Accredited by NBA, New Delhi, NAAC A+

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ABSTRACT

Child vaccination plays a crucial role in protecting infants and toddlers from preventable diseases and ensuring healthy development. However, in many rural areas, vaccination records are still maintained using manual paper-based methods, which often leads to loss of data, missed vaccination schedules, and poor monitoring of immunization status. Parents may forget vaccination dates due to lack of reminders, and healthcare workers face difficulties in maintaining accurate records for multiple children. Additionally, many parents lack proper knowledge regarding age-based nutrition and diet planning for infants and toddlers.

This project proposes the development of a Smart Child Vaccination and Health Monitoring System using the Django web framework. The system provides a digital platform for registering child details, maintaining vaccination records, tracking upcoming vaccination schedules, and generating digital vaccination certificates automatically. It also includes a simple rule-based diet recommendation module that suggests suitable meals based on the child's age, helping parents provide proper nutrition using locally available resources.

The proposed system aims to improve vaccination tracking efficiency, reduce missed immunizations, enhance record accuracy, and support child health awareness in rural communities. By integrating healthcare record management and basic nutritional support into a single web-based system, this project demonstrates the effective use of full-stack development technologies in solving real-world healthcare challenges.

INTRODUCTION

Vaccination is one of the most effective public health interventions for preventing infectious diseases and improving child survival rates. Immunization protects children from diseases such as tuberculosis, measles, polio, and diphtheria, which can cause serious health complications if not prevented at an early stage. In developing regions and rural communities, maintaining proper vaccination schedules is essential for ensuring the well-being of children.

Despite government initiatives to promote vaccination programs, many rural healthcare centers still depend on manual record-keeping methods. These paper-based records are prone to damage, loss, and human error. Parents often forget scheduled vaccination dates due to lack of proper reminders, resulting in delayed or missed immunizations. Healthcare staff also face challenges in tracking vaccination histories for multiple children efficiently.

With the advancement of digital technologies, web-based systems provide an efficient solution for managing healthcare records. Full-stack development frameworks such as Django allow developers to create secure, scalable, and user-friendly applications that can store large amounts of data and automate repetitive tasks. Digital systems can generate alerts, maintain accurate records, and provide useful recommendations to users.

This project focuses on developing a Smart Child Vaccination and Health Monitoring System using Django as the backend framework. The system allows healthcare staff to register children, update vaccination details, and monitor vaccination schedules in real time. It also generates digital vaccination certificates automatically, reducing paperwork and improving record reliability.

In addition to vaccination tracking, the system includes a simple diet recommendation module that suggests age-based nutritional meals for infants and toddlers. Proper nutrition during early childhood is essential for physical growth and brain development. Providing simple meal suggestions helps parents ensure that children receive balanced nutrition using commonly available foods.

The proposed system aims to simplify healthcare record management, reduce missed vaccinations, improve parental awareness, and support overall child health through a centralized digital platform.

PROBLEM STATEMENT

In many rural healthcare centers, vaccination records are maintained manually using paper-based systems. These manual methods often lead to several problems, such as loss of records, incorrect data entry, and difficulty in retrieving vaccination histories. Parents frequently forget vaccination dates due to lack of proper reminders, which results in incomplete immunization and increased risk of preventable diseases.

Healthcare workers also face challenges in managing large volumes of data manually, especially when handling records for multiple children. Generating vaccination certificates manually consumes time and increases the chances of human error. Additionally, there is limited awareness among parents regarding proper nutrition and diet planning for infants and toddlers, which can affect children's physical growth and cognitive development.

Existing digital systems mainly focus on vaccination tracking but often lack integrated features such as certificate generation and nutritional guidance. Therefore, there is a need for an integrated digital platform that can manage vaccination records efficiently, provide timely alerts for upcoming vaccinations, generate digital certificates automatically, and assist parents with simple diet recommendations.

The proposed system addresses these challenges by developing a web-based Smart Child Vaccination and Health Monitoring System using Django, which improves record accuracy, reduces missed vaccinations, and supports child health management.

OBJECTIVES

The main objectives of the proposed system are:

1. To design and develop a web-based system for registering child details in a structured digital format.
2. To maintain accurate digital vaccination records for each registered child and ensure secure storage of data.
3. To track vaccination schedules based on the child's date of birth and automatically display upcoming vaccination dates.
4. To provide alerts for upcoming and missed vaccinations to ensure timely immunization.
5. To generate digital vaccination certificates automatically after completion of each vaccination.
6. To develop a simple rule-based diet recommendation module that suggests age-based meal plans for infants and toddlers.
7. To create a user-friendly interface that can be easily used by healthcare staff and parents.
8. To improve awareness about vaccination schedules and child nutrition in rural communities.
9. To reduce manual paperwork and improve efficiency in healthcare record management.

LITERATURE SURVEY

Various researchers have developed systems to improve child healthcare through vaccination tracking, nutrition recommendation, and health monitoring technologies.

1. M. Zope, A. Suryawanshi, and S. De (2025) proposed an *Automated Vaccination Reminder and Tracking System for Rural Areas*. Their system focuses on sending automated reminders to parents about upcoming vaccination schedules using mobile technology. The study highlights how timely notifications significantly improve vaccination coverage in rural regions where awareness and access to healthcare services are limited.

2. A. B. Sallow, S. R. M. Zeebaree, and R. R. Zebari (2020) developed a *Vaccine Tracker/SMS Reminder System*. The system uses SMS-based alerts to notify parents about vaccination dates and maintain digital vaccination records. Their work demonstrates that automated reminders reduce missed vaccinations and improve record management efficiency.

3. I. Perumal and N. Ponnusamy (2025) introduced an *Intelligent Vaccination Reminder System* using a cross-stack development methodology. Their system integrates multiple technologies to provide reminders, record storage, and user-friendly interfaces. The study emphasizes the importance of integrating web-based platforms to simplify vaccination management for caregivers.

4. Z. A. Mekonnen, F. N. Hussien, and B. Tilahun (2019) designed an *Automated Text-Message Reminder System* to improve child vaccination uptake. Their research shows that text-message reminders significantly increase attendance at vaccination centers and reduce the risk of missed immunization schedules.

5. U. M. Umoren, B. A. Ojokoh, and O. S. Ijarotimi (2022) developed a *Diet Recommender System for School-Aged Children*. The system uses collaborative filtering and expert knowledge to recommend suitable diets based on children's nutritional needs. Their work focuses on reducing malnutrition and promoting healthy growth among children.

6. M. Hazman and A. M. Idrees (2015) proposed a *Healthy Nutrition Expert System for Children*. This system provides personalized dietary recommendations based on health conditions and age groups. The study highlights the importance of expert-based systems in improving children's nutritional habits and supporting healthy development.

7. G. S. Fuentes and G. L. D. Intal (2020) developed an *E-healthcare Child Monitoring Health System (CHMS) with SMS Functionality*. The system enables healthcare providers to monitor children's health records and send SMS alerts to parents. Their research demonstrates the usefulness of digital health monitoring in improving healthcare accessibility and efficiently.

PROPOSED SYSTEM

The proposed system, titled **Smart Child Vaccination and Health Monitoring System**, is a web-based application developed using the **Django framework** to improve the management of child healthcare records in rural and government healthcare environments. The system replaces traditional paper-based records with a centralized digital platform that enables efficient tracking of vaccination schedules, child health details, and nutrition support.

In this system, healthcare staff such as hospital workers can register children by entering essential details including the child's name, date of birth, parent information, contact number, and village details. After registration, the system automatically generates a vaccination schedule based on the child's age and recommended immunization timeline. All vaccination records are securely stored in a digital database, ensuring easy access and reducing the chances of data loss.

A key feature of the system is the **vaccination tracking and alert mechanism**, which monitors upcoming vaccination dates and generates reminders. If a vaccination is missed, the system provides notifications to healthcare staff and parents, helping to ensure timely immunization and reduce missed doses.

The system also includes **automatic digital certificate generation**, where a vaccination certificate is created in PDF format after each completed vaccine. The certificate contains essential details such as child name, vaccine name, vaccination date, and healthcare center information, serving as official proof of vaccination.

Additionally, the proposed system includes a **nutrition recommendation module** that suggests age-appropriate meals for infants and toddlers using commonly available food items. It may also provide simple recipes to support healthy growth and nutrition.

The system further supports **child health monitoring** by allowing healthcare workers to record growth details such as weight, height, and developmental milestones. It can also suggest age-appropriate activities that support brain development and early learning.

Overall, the system is designed with a **user-friendly interface** to ensure ease of use for healthcare staff and parents. By integrating vaccination tracking, diet recommendation, and child health monitoring into a single platform, the proposed system aims to improve healthcare services and promote better child development in rural communities.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements
The hardware components required for developing and running the system include:

- Personal Computer or Laptop
- Minimum 4 GB RAM (8 GB recommended)
- Processor: Intel i3 or above
- Hard Disk: Minimum 20 GB free space
- Internet Connection
- Printer (optional, for printing certificates)

These hardware components are sufficient for developing and testing the web-based system efficiently.

Software Requirements
The software tools required for developing the system include:
Operating System:

- Windows 10 or above
  OR
- Linux / macOS

Programming Language:

- Python

Web Framework:

- Django

Frontend Technologies:

- HTML
- CSS
- JavaScript
- Bootstrap

Database:

- SQLite (Default Django Database)

Development Tools:

- Visual Studio Code
- Web Browser (Google Chrome recommended)

Libraries Used:

- PDF generation library (for certificate creation)

These software tools provide a complete environment for developing and testing the proposed system.

METHODOLOGY

The development of the **Smart Child Vaccination and Health Monitoring System** follows a structured approach to design, develop, and implement the web-based application using the Django framework

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1. Requirement Analysis

In this phase, the requirements of the system were identified by studying the limitations of traditional paper-based vaccination records, such as difficulty in tracking schedules, risk of record loss, and lack of timely reminders. Based on this analysis, key features such as child registration, vaccination tracking, notification alerts, certificate generation, diet recommendation, and health monitoring were finalized.

2. System Design

During the design phase, the system architecture and modules were planned. The major modules include child registration, vaccination management, notification alerts, certificate generation, diet recommendation, and health monitoring. A database structure was designed to securely store child details, vaccination records, and health information.

3. System Development

The system was developed using the **Django framework** with **HTML, CSS, and JavaScript** for the user interface. Python was used for backend processing, and a database such as **SQLite/MySQL** was used to store records securely. All modules were developed and integrated to form a complete working system.

4. Testing

The system was tested to ensure correct functioning of all modules. Unit testing and system testing were performed to identify and fix errors, ensuring reliability and smooth performance.

5. Implementation and Maintenance

After testing, the system was implemented in a simulated healthcare environment. Regular maintenance and updates can be carried out to improve performance, fix issues, and add new features when required.

CONCLUSION

The **Smart Child Vaccination and Health Monitoring System** provides an efficient and reliable digital solution for managing child vaccination records and supporting healthcare activities. By replacing traditional paper-based methods with a web-based platform, the system improves data accuracy, minimizes the chances of record loss, and simplifies the overall process of vaccination tracking and management. The centralized storage of records also enables quick retrieval of information, reducing the time and effort required for maintaining child health data.

The integration of **alert notifications** ensures that parents and healthcare staff are regularly informed about upcoming and missed vaccinations. This feature helps maintain complete immunization schedules and reduces the risk of delayed or missed vaccinations, thereby improving overall child healthcare outcomes. The **digital certificate generation** feature further reduces manual paperwork and enhances documentation efficiency by providing instant vaccination proof in an organized digital format.

Additionally, the inclusion of a **diet recommendation module** supports child health by offering simple nutritional suggestions based on the child's age group. This promotes awareness among parents about proper feeding practices and encourages balanced nutrition during early childhood, which is essential for healthy growth and development.

Overall, the proposed system demonstrates how **full-stack development technologies**, particularly web-based frameworks, can be effectively applied to solve real-world healthcare challenges. The project highlights the importance of digital transformation in healthcare management, especially in rural and semi-urban areas where manual systems are still widely used. By improving vaccination tracking, documentation, and health awareness, the system contributes to better child health monitoring and supports the goal of ensuring timely and complete immunization for children.